

Some Electric Vehicles

Electric Cars in India

Mahindra e2o Plus: Starts at ₹ 730,000
Mahindra eVerito: Starts at ₹ 1.02 million
Hyundai Kona: Starts at ₹ 2.3 million

Expected Launches

Tata Tiago electric: Estimated price is ₹ 1.05 million (anticipated launch in July 2019)
Tata Tigor electric: Estimated price is ₹ 1.4 million (only available for government purchase; expected to be made available for public later this year)
Mahindra e-KUV 100: Estimate price unknown (anticipated launch later this year)
Audi e-tron: Estimated price ₹ 15 million (anticipated launch towards the end of 2019)

Some Hybrid Electric Cars in India

Toyota Camry: ₹ 3.69 million approximately
Toyota Prius: ₹ 4.5 million approximately
Honda Accord Hybrid: ₹ 4.32 million approximately
Lexus ES300h: ₹ 5.99 million approximately
Lexus NX300h SUV: ₹ 5.67 million approximately
Lexus RX450h: ₹ 12.9 million approximately
Lexus LS500h: ₹ 18.2 million approximately

Some Popular Electric Cars Globally

Kia Soul EV: US\$ 34,000
BMW i3: US\$ 44,500
Nissan Leaf: US\$ 30,000
Hyundai Ioniq: US\$ 29,500
Chevrolet Bolt: US\$ 36,500
Tesla Model S: US\$ 74,500

Note. This data has been collected based on data available online and on vendor sites.

EVs—e2o Plus in hatchback category, eVerito in sedan category and eSupro in cargo van category. These EVs fall in the more affordable price range of around one million rupees.

Mahindra is pushing the boundaries of EVs at a global scale, too. During Geneva Motor Show earlier this year, Pininfarina, a Mahindra company that makes luxury cars, unveiled its latest EV, named Battista, which is claimed to be the fastest EV, with an acceleration of 0mph to 62mph in two seconds. Battista will be available commercially in 2020.

Toyota offers two hybrid electric car models, namely, Prius and Camry.

These cars pack some advanced in-car features alongside their hybrid infrastructure, setting the price of the vehicles in the mid segment, of over ₹ 3.5 million.

Lexus has its own range of multiple hybrid electric offerings, three of which are ES300h in the higher price range (over ₹ 5 million) and LS500h and RX450h in the premium price range (over ₹ 10 million).

Tata Motors is all set to come out with its electric cars, namely, Tiago electric, Tigor electric and Nano electric. Tiago will be available for purchase soon. While Tigor is being sold for government purposes only, Tata will be opening it up to the public later this year. Nano electric is also towards the front of the pipeline. All of these EVs will be available at affordable prices, too.

Players like Audi and BMW are also planning to launch their EVs soon. Their offerings, as expected, will fall in the luxury price range.



Mahindra e2o Plus

Performance of EVs

EV makers are gradually improving their designs and configurations to match with traditional petrol or diesel cars. There is still much ground to be covered, though. Medium-priced EVs in India can run for an average of 130 kilometres on a fully-charged battery. Costlier vehicles provide higher mileage. Depending on battery capacity—usually within the range of 11kWh - 18kWh—EVs can take between five to eight hours to fully recharge the battery using 230V AC supply.

EVs that come with an electronics-assisted braking system (EABS) consume the energy generated when the brake is applied, and transfer and store the energy in the battery, lowering its exhaustion rate as well as wastage of energy. Many cars come with a feature called revive mode, which can provide an additional eight to twelve kilometres after the battery has been exhausted. Maximum speed of EVs in India is much lower than their fuel-driven counterparts. Top speed is around 80kmph to 100kmph.

Charging infrastructure

Public infrastructure in India, like gas stations or parking lots, are slowly building facilities for charging EVs. Some gas stations are also bringing fast-charging infrastructure that would charge the battery within 50 to 90 minutes. However, the number of these facilities is low today, which is one of the main concerns for EV owners.

Lewis says, "One has to understand that charging is largely going to be done at home. It is the rare and one-off charging that will be required on road, which we call crisis charging. To that end, there are limited charging points both for residential and on-road charging. Even for residential charging, one has to understand the challenges, since in India we have apartment-based lifestyles instead of unit houses as in Europe or the Americas."

So, while individual house-cum-garage owners can ensure they have electric ports for EV charging lined with their own electric meters, those dwelling in apartments need to arrange for charging points at their